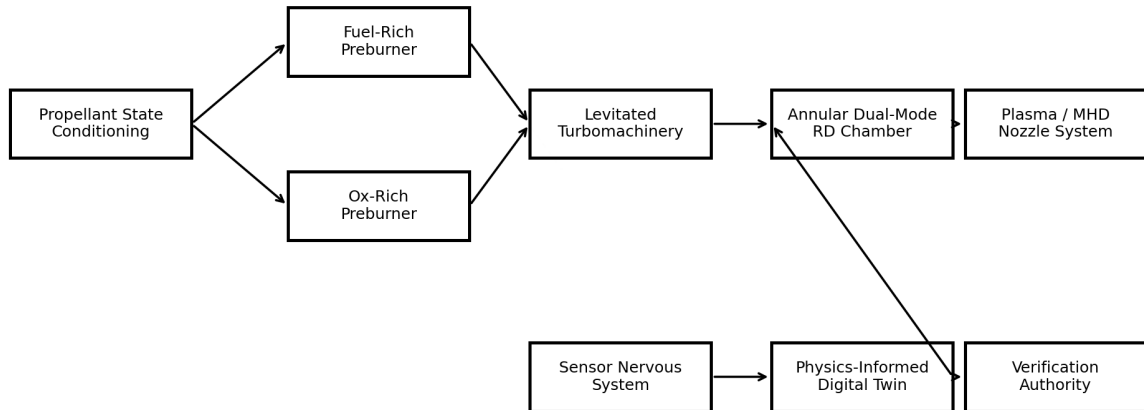


Chimera Prometheus

Implementation Manual Appendix

Paper-Only Multiphysics Systems Engineering Edition

Appendix Figure A-1: Non-operational systems architecture for Chimera Prometheus



A speculative, non-operational appendix for implementing the *model*, *requirements*, *simulation stack*, and *verification philosophy* of a post-Raptor conceptual engine.

Safety scope: This appendix is not a fabrication manual, test procedure, procurement guide, propellant-handling guide, or hot-fire checklist. It is a systems-engineering and mathematical implementation guide for a fictional/philosophical research artifact. It deliberately omits dimensions, manufacturing recipes, operating sequences, valve timings, fueling procedures, and actionable construction data.

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1 Purpose, Scope, and Safety Boundary

Boundary note. The phrase “implementation manual” is used here in the model-based systems engineering sense. The objective is to implement the concept as a traceable research program, a computational model, a verification matrix, and a coherent design dossier. Nothing in this appendix should be interpreted as instruction to build, acquire, operate, ignite, fuel, pressurize, or test rocket hardware.

This appendix converts Chimera Prometheus from a visionary manifesto into a disciplined paper architecture. The engine remains intentionally futuristic: a full-flow staged-combustion, rotating-detonation, plasma-assisted, magnetically levitated, self-sensing methane/oxygen propulsion concept. The implementation target is a corpus of analysis artifacts: requirements, equations, digital twin modules, safety claims, interface definitions, risk matrices, paper test articles, and research gates.

The guiding principle is **non-operational rigor**. Every subsystem is described strongly enough that a reader can understand the physics and the cross-couplings, but weakly enough that the document cannot serve as a build manual. The deliverable is therefore a “PhD appendix” rather than a shop drawing set.

1.1 The Four Implementation Products

1. **Mathematical specification:** governing equations, state variables, conserved quantities, closure assumptions, and nondimensional groups.
2. **Model architecture:** reduced-order models, high-fidelity submodels, uncertainty propagation, and digital twin update logic.
3. **Requirements and interfaces:** traceable performance, stability, observability, and credibility requirements.
4. **Verification dossier:** model review gates, hazard closure, FMEA, and evidence taxonomy.

1.2 Forbidden Content Classes

This appendix excludes operational details such as propellant loading, exact mixture schedules, startup sequences, pressurization procedure, injector hole sizing, nozzle dimensions, valve timing, turbopump blade geometry, material heat-treatment recipes, and any test-stand operating procedure. Where such details would normally appear, this appendix substitutes abstract variables, nondimensional constraints, and review criteria.

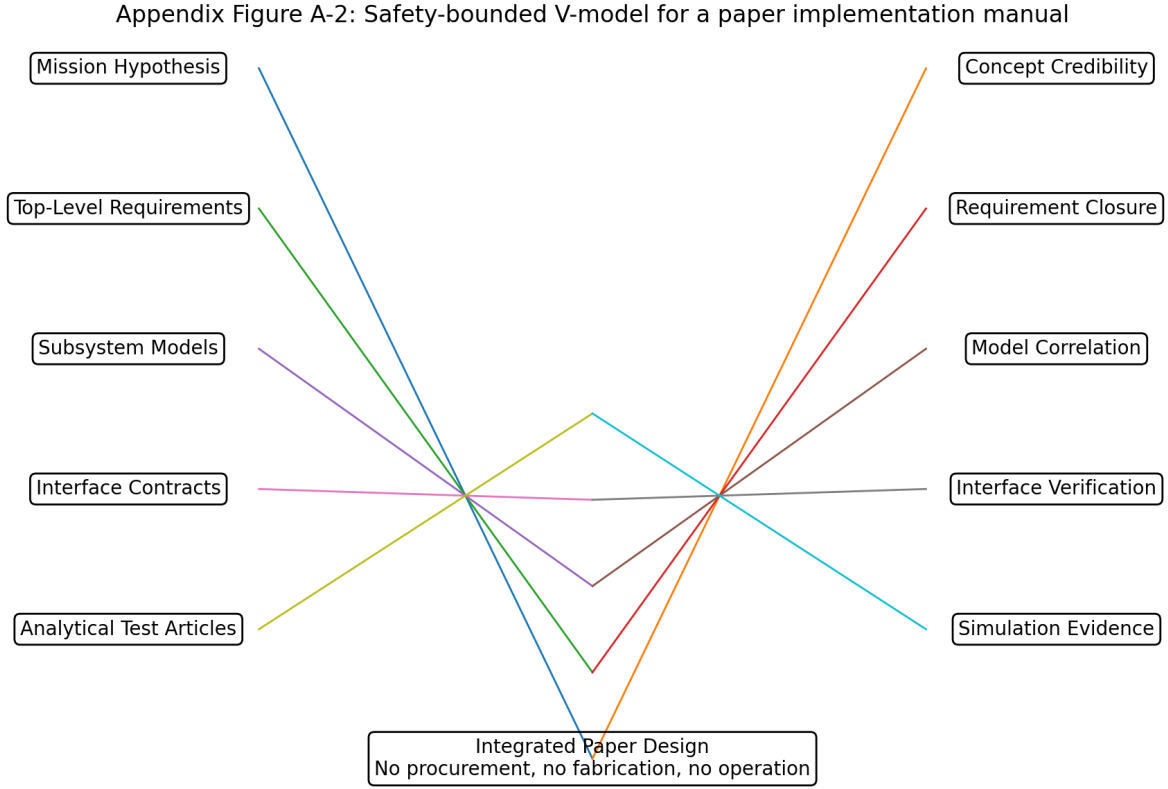


Figure 1: Implementation means closure of a paper V-model, not physical operation.

2 Top-Level Architecture and Requirements

2.1 Conceptual Architecture

The Chimera Prometheus architecture is partitioned into seven paper subsystems:

1. Propellant thermodynamic state conditioning.
2. Dual preburner energy conversion.
3. Contactless, counter-rotating turbomachinery.
4. Annular dual-mode deflagration/detonation chamber.
5. Hybrid regenerative, transpiration, and plasma-sheath thermal protection.
6. Magnetohydrodynamic nozzle and virtual expansion control.
7. Sensor nervous system and physics-informed digital twin.

A compact state vector for the integrated paper model is

$$\mathbf{x} = [P_c \quad T_c \quad \phi_c \quad \dot{m}_f \quad \dot{m}_o \quad \omega_f \quad \omega_o \quad \mathbf{a}_p \quad \mathbf{q}_w \quad \mathbf{z}_{RD} \quad \mathbf{b}_{MHD} \quad \boldsymbol{\theta}_{mat}]^T, \quad (1)$$

where P_c is chamber pressure, T_c is chamber temperature, ϕ_c is equivalence ratio, ω are rotor speeds, $\mathbf{a}_p$ are acoustic modal amplitudes, $\mathbf{q}_w$ are wall heat-flux states, $\mathbf{z}_{RD}$ describes detonation-wave phase and amplitude, $\mathbf{b}_{MHD}$ contains electromagnetic-field variables, and $\boldsymbol{\theta}_{mat}$ represents evolving material state.

2.2 Requirement Families

Family	Paper requirement intent
Performance	Establish symbolic relations for thrust, impulse, pressure-gain potential, and expansion control without prescribing hardware dimensions.
Stability	Require bounded acoustic, detonation, rotor, and control modes in analysis.
Thermal	Require every hot-wall node to close a conservative energy balance with explicit uncertainty.
Materials	Require redox compatibility, thermal fatigue logic, manufacturability evidence, and inspection paths.
Observability	Require each hidden state to be estimable by at least one theoretical sensor class or model inference path.
Safety	Require all paper studies to remain non-operational and to reject test procedures or build recipes.

2.3 System Objective Function

A paper optimization problem can be written as

$$\max_{\mathbf{u}, \boldsymbol{\theta}} J = w_I \hat{I}_{sp} + w_\eta \eta_{ex} + w_S S_{spectacle} + w_R R_{research} - w_H H_{hazard} - w_C C_{complexity}, \quad (2)$$

subject to

$$\mathbf{g}(\mathbf{x}, \mathbf{u}, \boldsymbol{\theta}) \leq \mathbf{0}, \quad (3)$$

$$\mathbf{h}(\mathbf{x}, \mathbf{u}, \boldsymbol{\theta}) = \mathbf{0}, \quad (4)$$

$$\mathcal{B}_{safe}(\mathbf{x}, \mathbf{u}) = 1. \quad (5)$$

Here $S_{spectacle}$ intentionally represents the visual and intellectual grandeur of the concept, while $\mathcal{B}_{safe}$ is a binary documentation boundary that rejects operational specificity.

3 Reference Equations for the Paper Cycle Model

3.1 Conservation Structure

The core cycle model is organized around mass, species, energy, momentum, entropy, and electromagnetic energy balances. For any abstract control volume Ω with boundary $\partial\Omega$,

$$\frac{d}{dt} \int_{\Omega} \rho dV + \int_{\partial\Omega} \rho \mathbf{v} \cdot \mathbf{n} dA = 0. \quad (6)$$

Species transport is written as

$$\frac{\partial(\rho Y_k)}{\partial t} + \nabla \cdot (\rho Y_k \mathbf{v}) = -\nabla \cdot \mathbf{J}_k + \dot{\omega}_k, \quad (7)$$

where Y_k is species mass fraction, $\mathbf{J}_k$ is diffusive flux, and $\dot{\omega}_k$ is the chemical source term.

The total energy equation, with electromagnetic work and radiative exchange retained symbolically, is

$$\frac{\partial(\rho E)}{\partial t} + \nabla \cdot [(\rho E + p) \mathbf{v}] = \nabla \cdot (k \nabla T) + \nabla \cdot (\boldsymbol{\tau} \cdot \mathbf{v}) + \dot{q}_{chem} + \dot{q}_{rad} + \mathbf{J} \cdot \mathbf{E}. \quad (8)$$

3.2 Thrust and Effective Exhaust Velocity

For the paper engine,

$$F = \dot{m} v_e + (P_e - P_a) A_e + F_{MHD} + F_{plasma}, \quad (9)$$

where F_{MHD} and F_{plasma} are not assumed to be large; they are accounting terms for electromagnetic and plasma-mediated momentum manipulation. The effective specific impulse is

$$I_{sp,eff} = \frac{F}{\dot{m} g_0}. \quad (10)$$

3.3 Pressure-Gain Combustion Metric

A rotating detonation chamber is assigned a pressure-gain factor

$$\Pi_{PGC} = \frac{\langle P_{t,out} \rangle}{\langle P_{t,in} \rangle}, \quad (11)$$

where P_t denotes total pressure. Conventional deflagration typically incurs $\Pi_{PGC} < 1$; the Prometheus ideal seeks symbolic regimes where $\Pi_{PGC} > 1$ while stability metrics remain bounded.

3.4 Cycle Closure

The abstract power-balance closure for each shaft is

$$\eta_t \dot{m}_g c_p T_{t,in} \left[1 - \left(\frac{P_{t,out}}{P_{t,in}} \right)^{(\gamma-1)/\gamma} \right] = \frac{\dot{m}_\ell \Delta P}{\rho \ell \eta_p} + P_{bearing} + P_{EM} + P_{margin}. \quad (12)$$

This is a paper equation only; no numeric pump sizing is supplied.

4 Dual-Mode Rotating Detonation Chamber Implementation

4.1 State Representation

The rotating detonation module is treated as an annular reacting-flow oscillator. Its reduced state can be written as

$$\mathbf{z}_{RD} = [N_w \ \psi_1 \ \dots \ \psi_{N_w} \ A_1 \ \dots \ A_{N_w} \ \lambda_m \ \chi_q]^T, \quad (13)$$

where N_w is wave count, ψ_i are wave phases, A_i are normalized wave amplitudes, λ_m is a mixing completion variable, and χ_q is a heat-release localization parameter.

4.2 Wave Kinematics

For an annulus of symbolic radius R_a , the azimuthal wave speed relation is

$$\dot{\psi}_i = \frac{D_i}{R_a} + \Delta\omega_i(\mathbf{x}), \quad (14)$$

where D_i is a detonation-wave speed surrogate and $\Delta\omega_i$ accounts for injector coupling, wall loss, curvature, and nonuniform mixture state.

A non-dimensional detonation viability parameter is

$$\mathcal{D} = \frac{\tau_{mix}}{\tau_{chem}} \frac{L_{res}}{\lambda_{cell}} \frac{P_t}{P_{ref}} \Theta_{loss}^{-1}, \quad (15)$$

where τ_{mix} is mixing time, τ_{chem} is chemical time, L_{res} is residence length, λ_{cell} is a detonation-cell surrogate, and Θ_{loss} represents wall and geometric losses. The paper rule is that detonation mode may be analyzed only when

$$\mathcal{D} > \mathcal{D}_{crit} \quad \text{and} \quad \sigma(\mathbf{a}_p) < \sigma_{max}. \quad (16)$$

4.3 Mode Transition Logic

The concept uses mode labels, not operational steps:

$$\mathcal{M} \in \{\text{soft, deflagration, assist, rotating detonation, plasma trim}\}. \quad (17)$$

A supervisory automaton may be specified as

$$\mathcal{M}_{k+1} = \Gamma(\mathcal{M}_k, \mathbf{x}_k, \hat{\mathbf{x}}_k, \mathcal{R}_k), \quad (18)$$

where Γ is a formal transition map and $\mathcal{R}_k$ is the current risk-state vector. The appendix deliberately does not define time sequences, valve trajectories, or ignition commands.

5 Thermal Protection and Wall-Stack Implementation

Appendix Figure A-6: Multiphysics wall-stack concept

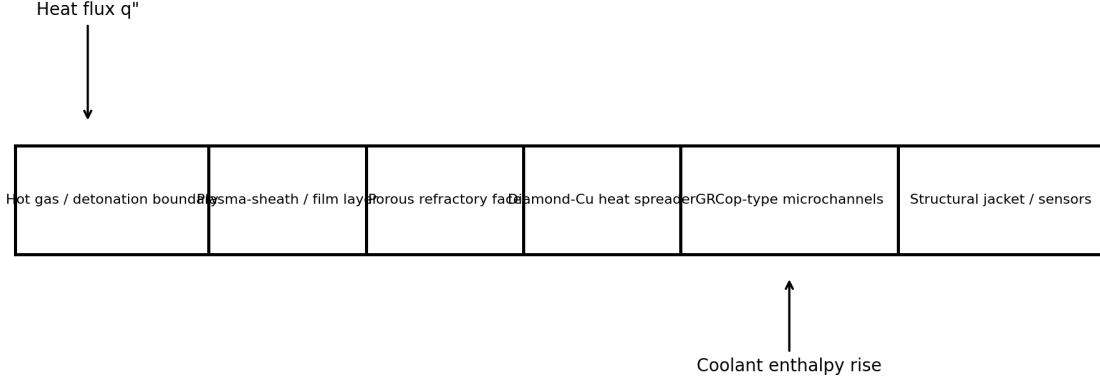


Figure 2: Layered thermal implementation: a symbolic wall stack for paper analysis only.

5.1 Conjugate Heat Transfer

For a wall coordinate y measured from the hot side, the one-dimensional reduced model is

$$\rho_w c_w \frac{\partial T_w}{\partial t} = \frac{\partial}{\partial y} \left(k_w(T) \frac{\partial T_w}{\partial y} \right) + \dot{q}_{vol}(y, t), \quad (19)$$

with boundary condition

$$-k_w \frac{\partial T_w}{\partial y} \Big|_{hot} = q''_{conv} + q''_{rad} - q''_{film} - q''_{plasma}. \quad (20)$$

The coolant-side balance is

$$\dot{m}_{cool} c_{p,c} \frac{dT_c}{ds} = h_c P_{wet} (T_w - T_c) - \dot{q}_{endo}, \quad (21)$$

where $\dot{q}_{endo}$ represents optional endothermic fuel-conditioning chemistry, retained only as a theoretical heat-sink term.

5.2 Transpiration Layer

The porous layer is represented by Darcy-Forchheimer flow:

$$-\nabla p = \frac{\mu}{K} \mathbf{u} + \rho C_F |\mathbf{u}| \mathbf{u}, \quad (22)$$

where K is permeability and C_F is an inertial coefficient. The protective effectiveness is described by

$$\eta_{tr} = 1 - \frac{q''_{wall,tr}}{q''_{wall,0}}, \quad (23)$$

subject to the qualitative constraint that no pore-scale prescription is provided.

5.3 Thermal Runaway Index

A paper thermal runaway index is

$$\mathcal{T}_{run} = \max_y \left[\frac{T_w(y) - T_{allow}(y)}{T_{allow}(y) - T_{ref}} \right] + \alpha \|\nabla T_w\|_2 + \beta \left\| \frac{dT_w}{dt} \right\|_2. \quad (24)$$

A valid paper configuration requires $\mathcal{T}_{run} < 0$ for all credible analytic load cases.

6 Magnetohydrodynamic Nozzle and Plasma Sheath

6.1 MHD Coupling

The plasma-assisted nozzle is governed symbolically by the resistive MHD equations. Maxwell's reduced low-frequency form is

$$\nabla \cdot \mathbf{B} = 0, \quad (25)$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}, \quad (26)$$

$$\mathbf{J} = \sigma_e (\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (27)$$

The momentum equation includes a Lorentz force:

$$\rho \frac{D\mathbf{v}}{Dt} = -\nabla p + \nabla \cdot \boldsymbol{\tau} + \mathbf{J} \times \mathbf{B}. \quad (28)$$

6.2 MHD Power Harvesting

The theoretical electrical power density is

$$\dot{w}_{MHD} = \mathbf{J} \cdot \mathbf{E}_{load}, \quad (29)$$

and an idealized channel efficiency may be written

$$\eta_{MHD} = \frac{\int_{\Omega_{MHD}} \mathbf{J} \cdot \mathbf{E}_{load} dV}{\dot{m} (h_0 - h_e)}. \quad (30)$$

The manual requires this term to be treated as speculative and bounded by uncertainty intervals.

6.3 Magnetic Virtual Nozzle

The effective expansion area is represented by

$$A_{e,eff} = A_{geom} + \Delta A_{plasma}(\mathbf{B}, \sigma_e, M, \chi_i), \quad (31)$$

where χ_i is ionization fraction. The paper claim is not that such a nozzle is practical, but that a model can parameterize how electromagnetic fields could perturb exhaust expansion.

7 Levitated Turbomachinery and Rotor Dynamics

7.1 Rotor Equations

Each rotor is modeled as a flexible gyroscopic system:

$$\mathbf{M}\ddot{\mathbf{q}} + (\mathbf{C} + \Omega\mathbf{G})\dot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{f}_{fluid} + \mathbf{f}_{mag} + \mathbf{f}_{imb} + \mathbf{f}_{ctrl}. \quad (32)$$

The state-space form is

$$\dot{\mathbf{x}}_r = \mathbf{A}(\Omega, \theta_b)\mathbf{x}_r + \mathbf{B}\mathbf{u}_b + \mathbf{w}_r, \quad (33)$$

where θ_b encodes bearing stiffness, damping, flux-pinning state, and thermal condition.

7.2 Flux-Pinning Surrogate

A passive superconducting restoring model may be represented as

$$\mathbf{f}_{pin} = -\mathbf{K}_{pin}(T, B, \Delta)\mathbf{q} - \mathbf{C}_{pin}(T, B, \Delta)\dot{\mathbf{q}}, \quad (34)$$

where Δ summarizes trapped-flux history. A paper safety criterion is

$$\lambda_{max}[\mathbf{A}_{closed}(\Omega)] < 0 \quad \forall \Omega \in \Omega_{paper}. \quad (35)$$

7.3 Motor-Generator Energy Accounting

The shaft electrical balance is

$$P_{elec} = \eta_g \tau_g \Omega - \frac{\tau_m \Omega}{\eta_m} - P_{ctrl} - P_{loss}, \quad (36)$$

with τ_g and τ_m never translated into construction data. The motor-generator is a paper mechanism for closing startup-energy, shutdown-energy, and sensor-power arguments.

8 Materials, Redox Zoning, and Manufacturing Evidence

8.1 Redox-Aware Materials Map

The material field is not uniform. Define a redox potential surrogate $\mathcal{R}_{ox}$ and a carburization potential $\mathcal{R}_{carb}$:

$$\mathcal{R}_{ox} = f(Y_{O_2}, T, p, \phi^{-1}), \quad (37)$$

$$\mathcal{R}_{carb} = g(Y_{CH_4}, Y_{CO}, Y_C, T, p, \phi). \quad (38)$$

A candidate material stack is admissible only if

$$\mathcal{M}_{allow}(\mathbf{r}) \in \arg \min_{m \in \mathcal{S}} [w_T \Pi_T + w_R \Pi_R + w_F \Pi_F + w_I \Pi_I], \quad (39)$$

where Π_T is thermal penalty, Π_R redox penalty, Π_F fatigue penalty, and Π_I inspectability penalty.

8.2 Functionally Graded Field

A continuous material composition vector $\mathbf{c}(\mathbf{r})$ obeys

$$\sum_i c_i(\mathbf{r}) = 1, \quad c_i \geq 0, \quad \|\nabla \mathbf{c}\| \leq \Lambda_{grad}. \quad (40)$$

The gradient constraint prevents imaginary discontinuities that would make the paper design internally inconsistent.

8.3 Additive Manufacturing Pedigree

For a paper AM implementation, every voxel v receives a pedigree vector

$$\mathbf{p}_v = [E_v \quad \dot{T}_v \quad \phi_v \quad \epsilon_{res,v} \quad d_{grain,v} \quad \xi_{defect,v}]^T, \quad (41)$$

where E_v is energy-density surrogate, $\dot{T}_v$ cooling-rate surrogate, ϕ_v porosity, $\epsilon_{res,v}$ residual strain, and $\xi_{defect,v}$ defect probability. The material digital twin maps

$$\boldsymbol{\theta}_{mat} = \mathcal{K}(\{\mathbf{p}_v\}, \mathcal{H}_{thermal}, \mathcal{H}_{stress}). \quad (42)$$

9 Sensor Nervous System and Observability

9.1 Measurement Model

The abstract sensor network returns

$$\mathbf{y} = \mathbf{H}(\mathbf{x}) + \mathbf{v}, \quad \mathbf{v} \sim \mathcal{N}(0, \mathbf{R}), \quad (43)$$

with sensor families

$$\mathbf{y} = [\mathbf{y}_{pressure} \quad \mathbf{y}_{optical} \quad \mathbf{y}_{fiber} \quad \mathbf{y}_{rotor} \quad \mathbf{y}_{thermal} \quad \mathbf{y}_{plasma} \quad \mathbf{y}_{chem}]^T. \quad (44)$$

9.2 Distributed Fiber Sensing

For fiber Bragg grating diagnostics,

$$\frac{\Delta\lambda_B}{\lambda_B} = (1 - p_e)\epsilon + (\alpha_f + \xi_f)\Delta T. \quad (45)$$

The paper implementation uses this equation only to justify strain-temperature separation in the model.

9.3 Spectroscopic Combustion State

Optical emission and absorption are reduced to a species inference map:

$$\hat{\mathbf{Y}} = \mathcal{S}^{-1}(I_{\lambda_1}, I_{\lambda_2}, \dots, I_{\lambda_n}; T, p, \ell), \quad (46)$$

where I_λ are line or band intensities and ℓ is path length. The mixture-ratio estimate becomes

$$\hat{O}/\hat{F} = \mathcal{C}(\hat{Y}_{CO_2}, \hat{Y}_{H_2O}, \hat{Y}_{CO}, \hat{Y}_{O_2}, \hat{Y}_{CH_4}). \quad (47)$$

9.4 Observability Criterion

The nonlinear observability matrix is

$$\mathcal{O} = \begin{bmatrix} \nabla L_f^0 h \\ \nabla L_f^1 h \\ \dots \\ \nabla L_f^{n-1} h \end{bmatrix}. \quad (48)$$

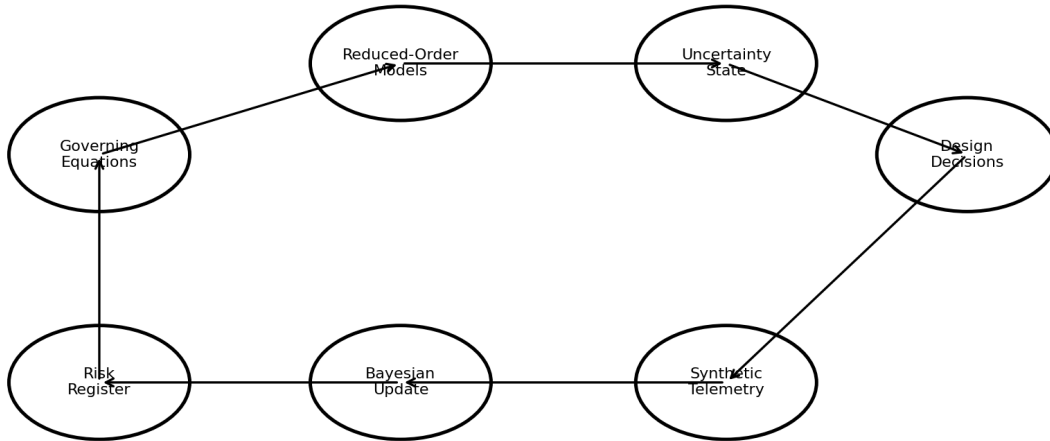
A paper design claim is admissible only where

$$\text{rank}(\mathcal{O}) \geq n_{critical}, \quad (49)$$

for the critical state subset.

10 Physics-Informed Digital Twin Implementation

Appendix Figure A-3: Physics-informed digital twin governance loop



The digital twin is a paper-research construct: it closes epistemic uncertainty, not hardware valves.

Figure 3: The digital twin governs claims, uncertainty, and review closure.

10.1 State Estimation

A general estimator is

$$\mathbf{x}_{k+1} = \mathbf{f}(\mathbf{x}_k, \mathbf{u}_k, \boldsymbol{\theta}) + \mathbf{w}_k, \quad (50)$$

$$\mathbf{y}_k = \mathbf{h}(\mathbf{x}_k) + \mathbf{v}_k. \quad (51)$$

For paper implementation, the preferred structure is ensemble-based:

$$\left\{ \mathbf{x}_k^{(j)} \right\}_{j=1}^{N_e} \rightarrow \left\{ \mathbf{x}_{k+1}^{(j)} \right\}_{j=1}^{N_e}, \quad (52)$$

with covariance

$$\mathbf{P} = \frac{1}{N_e - 1} \sum_{j=1}^{N_e} (\mathbf{x}^{(j)} - \bar{\mathbf{x}})(\mathbf{x}^{(j)} - \bar{\mathbf{x}})^T. \quad (53)$$

10.2 Bayesian Credibility Update

A model parameter vector $\boldsymbol{\theta}$ is updated by

$$p(\boldsymbol{\theta}|\mathcal{D}) = \frac{p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathcal{D})}, \quad (54)$$

where $\mathcal{D}$ may be synthetic analysis evidence, literature-derived bounding evidence, or non-operational surrogate data.

10.3 Physics-Informed Loss

A neural surrogate may use a loss

$$\mathcal{L} = \mathcal{L}_{data} + \lambda_m \mathcal{L}_{mass} + \lambda_E \mathcal{L}_{energy} + \lambda_s \mathcal{L}_{entropy} + \lambda_b \mathcal{L}_{boundary} + \lambda_{safe} \mathcal{L}_{scope}. \quad (55)$$

The final term penalizes outputs that resemble operational instructions.

11 Subsystem Implementation Cards

Each card defines how a subsystem is implemented as a paper artifact. The sections below are intentionally repetitive because a research appendix should be auditable: every subsystem receives a physics model, state variables, interfaces, evidence artifacts, and exclusion rules.

11.1 S1: Propellant State Conditioning

Physics themes: real-fluid thermodynamics, pseudo-boiling margins, entropy accounting, cryogenic heat exchange.

State subset.

$$\mathbf{x}_{S1} = [\xi_1 \quad \xi_2 \quad \xi_3 \quad \xi_4 \quad \xi_5]^T, \quad (56)$$

where the ξ_i are subsystem-specific nondimensional states: load fraction, thermal severity, stability margin, uncertainty level, and observability score.

Governing closure.

$$\mathbf{r}_{S1}(\mathbf{x}, \mathbf{u}, \boldsymbol{\theta}) = \mathbf{A}_{S1}\mathbf{x}_{S1} + \mathbf{B}_{S1}\mathbf{u}_{S1} + \boldsymbol{\Phi}_{S1}(\mathbf{x}) - \mathbf{b}_{S1} = \mathbf{0}. \quad (57)$$

The residual $\mathbf{r}_{S1}$ is the formal object that must be closed by analysis; it is not an instruction to fabricate hardware.

Interface contract.

$$\mathcal{I}_{S1} = [\dot{m} \quad P \quad T \quad h \quad s \quad \mathbf{a}_p \quad \mathbf{q}'' \quad \mathbf{y} \quad \Sigma]^T, \quad (58)$$

where Σ is the uncertainty covariance associated with the interface variables. Every interface must provide not only a nominal value but also uncertainty and validity range.

Acceptance logic.

$$M_{stability} > M_{min}, \quad (59)$$

$$\Pr(\mathcal{F}_{S1}) < \Pr_{max}, \quad (60)$$

$$\kappa(\mathcal{O}_{S1}) < \kappa_{max}, \quad (61)$$

$$\mathcal{E}_{evidence} \geq \mathcal{E}_{required}. \quad (62)$$

Here $\mathcal{F}$ is a failure proposition, $\mathcal{O}$ is the subsystem observability map, and κ is a conditioning metric.

Evidence artifacts. Required artifacts include a model note, an uncertainty note, a hazard note, a coupling note, and a verification note. Operational procedures, component dimensions, and procurement specifications are excluded.

11.2 S2: Fuel-Rich Preburner

Physics themes: rich-gas chemistry, soot/coking avoidance as a symbolic risk state, turbine compatibility.

State subset.

$$\mathbf{x}_{S2} = [\xi_1 \ \xi_2 \ \xi_3 \ \xi_4 \ \xi_5]^T, \quad (63)$$

where the ξ_i are subsystem-specific nondimensional states: load fraction, thermal severity, stability margin, uncertainty level, and observability score.

Governing closure.

$$\mathbf{r}_{S2}(\mathbf{x}, \mathbf{u}, \boldsymbol{\theta}) = \mathbf{A}_{S2}\mathbf{x}_{S2} + \mathbf{B}_{S2}\mathbf{u}_{S2} + \boldsymbol{\Phi}_{S2}(\mathbf{x}) - \mathbf{b}_{S2} = \mathbf{0}. \quad (64)$$

The residual $\mathbf{r}_{S2}$ is the formal object that must be closed by analysis; it is not an instruction to fabricate hardware.

Interface contract.

$$\mathcal{I}_{S2} = [\dot{m} \ P \ T \ h \ s \ \mathbf{a}_p \ \mathbf{q}'' \ \mathbf{y} \ \Sigma]^T, \quad (65)$$

where Σ is the uncertainty covariance associated with the interface variables. Every interface must provide not only a nominal value but also uncertainty and validity range.

Acceptance logic.

$$M_{stability} > M_{min}, \quad (66)$$

$$\Pr(\mathcal{F}_{S2}) < \Pr_{max}, \quad (67)$$

$$\kappa(\mathcal{O}_{S2}) < \kappa_{max}, \quad (68)$$

$$\mathcal{E}_{evidence} \geq \mathcal{E}_{required}. \quad (69)$$

Here $\mathcal{F}$ is a failure proposition, $\mathcal{O}$ is the subsystem observability map, and κ is a conditioning metric.

Evidence artifacts. Required artifacts include a model note, an uncertainty note, a hazard note, a coupling note, and a verification note. Operational procedures, component dimensions, and procurement specifications are excluded.

11.3 S3: Oxygen-Rich Preburner

Physics themes: redox severity, oxidizer compatibility, thermal margins, pressure oscillation filtering.

State subset.

$$\mathbf{x}_{S3} = [\xi_1 \ \xi_2 \ \xi_3 \ \xi_4 \ \xi_5]^T, \quad (70)$$

where the ξ_i are subsystem-specific nondimensional states: load fraction, thermal severity, stability margin, uncertainty level, and observability score.

Governing closure.

$$\mathbf{r}_{S3}(\mathbf{x}, \mathbf{u}, \boldsymbol{\theta}) = \mathbf{A}_{S3}\mathbf{x}_{S3} + \mathbf{B}_{S3}\mathbf{u}_{S3} + \boldsymbol{\Phi}_{S3}(\mathbf{x}) - \mathbf{b}_{S3} = \mathbf{0}. \quad (71)$$

The residual $\mathbf{r}_{S3}$ is the formal object that must be closed by analysis; it is not an instruction to fabricate hardware.

Interface contract.

$$\mathcal{I}_{S3} = [\dot{m} \ P \ T \ h \ s \ \mathbf{a}_p \ \mathbf{q}'' \ \mathbf{y} \ \Sigma]^T, \quad (72)$$

where Σ is the uncertainty covariance associated with the interface variables. Every interface must provide not only a nominal value but also uncertainty and validity range.

Acceptance logic.

$$M_{stability} > M_{min}, \quad (73)$$

$$\Pr(\mathcal{F}_{S3}) < \Pr_{max}, \quad (74)$$

$$\kappa(\mathcal{O}_{S3}) < \kappa_{max}, \quad (75)$$

$$\mathcal{E}_{evidence} \geq \mathcal{E}_{required}. \quad (76)$$

Here $\mathcal{F}$ is a failure proposition, $\mathcal{O}$ is the subsystem observability map, and κ is a conditioning metric.

Evidence artifacts. Required artifacts include a model note, an uncertainty note, a hazard note, a coupling note, and a verification note. Operational procedures, component dimensions, and procurement specifications are excluded.

11.4 S4: Counter-Rotating Turbopumps

Physics themes: rotor dynamics, active magnetic support, cavitation observability, generator coupling.

State subset.

$$\mathbf{x}_{S4} = [\xi_1 \ \xi_2 \ \xi_3 \ \xi_4 \ \xi_5]^T, \quad (77)$$

where the ξ_i are subsystem-specific nondimensional states: load fraction, thermal severity, stability margin, uncertainty level, and observability score.

Governing closure.

$$\mathbf{r}_{S4}(\mathbf{x}, \mathbf{u}, \boldsymbol{\theta}) = \mathbf{A}_{S4}\mathbf{x}_{S4} + \mathbf{B}_{S4}\mathbf{u}_{S4} + \boldsymbol{\Phi}_{S4}(\mathbf{x}) - \mathbf{b}_{S4} = \mathbf{0}. \quad (78)$$

The residual $\mathbf{r}_{S4}$ is the formal object that must be closed by analysis; it is not an instruction to fabricate hardware.

Interface contract.

$$\mathcal{I}_{S4} = [\dot{m} \ P \ T \ h \ s \ \mathbf{a}_p \ \mathbf{q}'' \ \mathbf{y} \ \Sigma]^T, \quad (79)$$

where Σ is the uncertainty covariance associated with the interface variables. Every interface must provide not only a nominal value but also uncertainty and validity range.

Acceptance logic.

$$M_{stability} > M_{min}, \quad (80)$$

$$\Pr(\mathcal{F}_{S4}) < \Pr_{max}, \quad (81)$$

$$\kappa(\mathcal{O}_{S4}) < \kappa_{max}, \quad (82)$$

$$\mathcal{E}_{evidence} \geq \mathcal{E}_{required}. \quad (83)$$

Here $\mathcal{F}$ is a failure proposition, $\mathcal{O}$ is the subsystem observability map, and κ is a conditioning metric.

Evidence artifacts. Required artifacts include a model note, an uncertainty note, a hazard note, a coupling note, and a verification note. Operational procedures, component dimensions, and procurement specifications are excluded.

11.5 S5: Annular RD Chamber

Physics themes: pressure-gain combustion, wave phasing, injector coupling, acoustic mode avoidance.

State subset.

$$\mathbf{x}_{S5} = [\xi_1 \ \xi_2 \ \xi_3 \ \xi_4 \ \xi_5]^T, \quad (84)$$

where the ξ_i are subsystem-specific nondimensional states: load fraction, thermal severity, stability margin, uncertainty level, and observability score.

Governing closure.

$$\mathbf{r}_{S5}(\mathbf{x}, \mathbf{u}, \boldsymbol{\theta}) = \mathbf{A}_{S5}\mathbf{x}_{S5} + \mathbf{B}_{S5}\mathbf{u}_{S5} + \boldsymbol{\Phi}_{S5}(\mathbf{x}) - \mathbf{b}_{S5} = \mathbf{0}. \quad (85)$$

The residual $\mathbf{r}_{S5}$ is the formal object that must be closed by analysis; it is not an instruction to fabricate hardware.

Interface contract.

$$\mathcal{I}_{S5} = [\dot{m} \ P \ T \ h \ s \ \mathbf{a}_p \ \mathbf{q}'' \ \mathbf{y} \ \Sigma]^T, \quad (86)$$

where Σ is the uncertainty covariance associated with the interface variables. Every interface must provide not only a nominal value but also uncertainty and validity range.

Acceptance logic.

$$M_{stability} > M_{min}, \quad (87)$$

$$\Pr(\mathcal{F}_{S5}) < \Pr_{max}, \quad (88)$$

$$\kappa(\mathcal{O}_{S5}) < \kappa_{max}, \quad (89)$$

$$\mathcal{E}_{evidence} \geq \mathcal{E}_{required}. \quad (90)$$

Here $\mathcal{F}$ is a failure proposition, $\mathcal{O}$ is the subsystem observability map, and κ is a conditioning metric.

Evidence artifacts. Required artifacts include a model note, an uncertainty note, a hazard note, a coupling note, and a verification note. Operational procedures, component dimensions, and procurement specifications are excluded.

11.6 S6: Adaptive Injector Face

Physics themes: modal injection shaping, mixture distribution estimation, phase-locked fueling theory.

State subset.

$$\mathbf{x}_{S6} = [\xi_1 \ \xi_2 \ \xi_3 \ \xi_4 \ \xi_5]^T, \quad (91)$$

where the ξ_i are subsystem-specific nondimensional states: load fraction, thermal severity, stability margin, uncertainty level, and observability score.

Governing closure.

$$\mathbf{r}_{S6}(\mathbf{x}, \mathbf{u}, \boldsymbol{\theta}) = \mathbf{A}_{S6}\mathbf{x}_{S6} + \mathbf{B}_{S6}\mathbf{u}_{S6} + \boldsymbol{\Phi}_{S6}(\mathbf{x}) - \mathbf{b}_{S6} = \mathbf{0}. \quad (92)$$

The residual $\mathbf{r}_{S6}$ is the formal object that must be closed by analysis; it is not an instruction to fabricate hardware.

Interface contract.

$$\mathcal{I}_{S6} = [\dot{m} \ P \ T \ h \ s \ \mathbf{a}_p \ \mathbf{q}'' \ \mathbf{y} \ \Sigma]^T, \quad (93)$$

where Σ is the uncertainty covariance associated with the interface variables. Every interface must provide not only a nominal value but also uncertainty and validity range.

Acceptance logic.

$$M_{stability} > M_{min}, \quad (94)$$

$$\Pr(\mathcal{F}_{S6}) < \Pr_{max}, \quad (95)$$

$$\kappa(\mathcal{O}_{S6}) < \kappa_{max}, \quad (96)$$

$$\mathcal{E}_{evidence} \geq \mathcal{E}_{required}. \quad (97)$$

Here $\mathcal{F}$ is a failure proposition, $\mathcal{O}$ is the subsystem observability map, and κ is a conditioning metric.

Evidence artifacts. Required artifacts include a model note, an uncertainty note, a hazard note, a coupling note, and a verification note. Operational procedures, component dimensions, and procurement specifications are excluded.

11.7 S7: Hybrid Cooling Network

Physics themes: conjugate heat transfer, porous flow, fractal coolant distribution, emergency heat absorption.

State subset.

$$\mathbf{x}_{S7} = [\xi_1 \ \xi_2 \ \xi_3 \ \xi_4 \ \xi_5]^T, \quad (98)$$

where the ξ_i are subsystem-specific nondimensional states: load fraction, thermal severity, stability margin, uncertainty level, and observability score.

Governing closure.

$$\mathbf{r}_{S7}(\mathbf{x}, \mathbf{u}, \boldsymbol{\theta}) = \mathbf{A}_{S7}\mathbf{x}_{S7} + \mathbf{B}_{S7}\mathbf{u}_{S7} + \boldsymbol{\Phi}_{S7}(\mathbf{x}) - \mathbf{b}_{S7} = \mathbf{0}. \quad (99)$$

The residual $\mathbf{r}_{S7}$ is the formal object that must be closed by analysis; it is not an instruction to fabricate hardware.

Interface contract.

$$\mathcal{I}_{S7} = [\dot{m} \ P \ T \ h \ s \ \mathbf{a}_p \ \mathbf{q}'' \ \mathbf{y} \ \Sigma]^T, \quad (100)$$

where Σ is the uncertainty covariance associated with the interface variables. Every interface must provide not only a nominal value but also uncertainty and validity range.

Acceptance logic.

$$M_{stability} > M_{min}, \quad (101)$$

$$\Pr(\mathcal{F}_{S7}) < \Pr_{max}, \quad (102)$$

$$\kappa(\mathcal{O}_{S7}) < \kappa_{max}, \quad (103)$$

$$\mathcal{E}_{evidence} \geq \mathcal{E}_{required}. \quad (104)$$

Here $\mathcal{F}$ is a failure proposition, $\mathcal{O}$ is the subsystem observability map, and κ is a conditioning metric.

Evidence artifacts. Required artifacts include a model note, an uncertainty note, a hazard note, a coupling note, and a verification note. Operational procedures, component dimensions, and procurement specifications are excluded.

11.8 S8: MHD Nozzle

Physics themes: Lorentz force accounting, plasma conductivity, virtual nozzle extension, power harvesting.

State subset.

$$\mathbf{x}_{S8} = [\xi_1 \ \xi_2 \ \xi_3 \ \xi_4 \ \xi_5]^T, \quad (105)$$

where the ξ_i are subsystem-specific nondimensional states: load fraction, thermal severity, stability margin, uncertainty level, and observability score.

Governing closure.

$$\mathbf{r}_{S8}(\mathbf{x}, \mathbf{u}, \boldsymbol{\theta}) = \mathbf{A}_{S8}\mathbf{x}_{S8} + \mathbf{B}_{S8}\mathbf{u}_{S8} + \boldsymbol{\Phi}_{S8}(\mathbf{x}) - \mathbf{b}_{S8} = \mathbf{0}. \quad (106)$$

The residual $\mathbf{r}_{S8}$ is the formal object that must be closed by analysis; it is not an instruction to fabricate hardware.

Interface contract.

$$\mathcal{I}_{S8} = [\dot{m} \ P \ T \ h \ s \ \mathbf{a}_p \ \mathbf{q}'' \ \mathbf{y} \ \Sigma]^T, \quad (107)$$

where Σ is the uncertainty covariance associated with the interface variables. Every interface must provide not only a nominal value but also uncertainty and validity range.

Acceptance logic.

$$M_{stability} > M_{min}, \quad (108)$$

$$\Pr(\mathcal{F}_{S8}) < \Pr_{max}, \quad (109)$$

$$\kappa(\mathcal{O}_{S8}) < \kappa_{max}, \quad (110)$$

$$\mathcal{E}_{evidence} \geq \mathcal{E}_{required}. \quad (111)$$

Here $\mathcal{F}$ is a failure proposition, $\mathcal{O}$ is the subsystem observability map, and κ is a conditioning metric.
Evidence artifacts. Required artifacts include a model note, an uncertainty note, a hazard note, a coupling note, and a verification note. Operational procedures, component dimensions, and procurement specifications are excluded.

11.9 S9: Acoustic Metamaterial Liner

Physics themes: Helmholtz lattices, phononic bandgaps, nonlinear damping, adaptive impedance.

State subset.

$$\mathbf{x}_{S9} = [\xi_1 \ \xi_2 \ \xi_3 \ \xi_4 \ \xi_5]^T, \quad (112)$$

where the ξ_i are subsystem-specific nondimensional states: load fraction, thermal severity, stability margin, uncertainty level, and observability score.

Governing closure.

$$\mathbf{r}_{S9}(\mathbf{x}, \mathbf{u}, \boldsymbol{\theta}) = \mathbf{A}_{S9}\mathbf{x}_{S9} + \mathbf{B}_{S9}\mathbf{u}_{S9} + \boldsymbol{\Phi}_{S9}(\mathbf{x}) - \mathbf{b}_{S9} = \mathbf{0}. \quad (113)$$

The residual $\mathbf{r}_{S9}$ is the formal object that must be closed by analysis; it is not an instruction to fabricate hardware.

Interface contract.

$$\mathcal{I}_{S9} = [\dot{m} \ P \ T \ h \ s \ \mathbf{a}_p \ \mathbf{q}'' \ \mathbf{y} \ \Sigma]^T, \quad (114)$$

where Σ is the uncertainty covariance associated with the interface variables. Every interface must provide not only a nominal value but also uncertainty and validity range.

Acceptance logic.

$$M_{stability} > M_{min}, \quad (115)$$

$$\Pr(\mathcal{F}_{S9}) < \Pr_{max}, \quad (116)$$

$$\kappa(\mathcal{O}_{S9}) < \kappa_{max}, \quad (117)$$

$$\mathcal{E}_{evidence} \geq \mathcal{E}_{required}. \quad (118)$$

Here $\mathcal{F}$ is a failure proposition, $\mathcal{O}$ is the subsystem observability map, and κ is a conditioning metric.

Evidence artifacts. Required artifacts include a model note, an uncertainty note, a hazard note, a coupling note, and a verification note. Operational procedures, component dimensions, and procurement specifications are excluded.

11.10 S10: Materials Digital Thread

Physics themes: voxel pedigree, redox map, fatigue accumulation, inspection closure.

State subset.

$$\mathbf{x}_{S10} = [\xi_1 \ \xi_2 \ \xi_3 \ \xi_4 \ \xi_5]^T, \quad (119)$$

where the ξ_i are subsystem-specific nondimensional states: load fraction, thermal severity, stability margin, uncertainty level, and observability score.

Governing closure.

$$\mathbf{r}_{S10}(\mathbf{x}, \mathbf{u}, \boldsymbol{\theta}) = \mathbf{A}_{S10}\mathbf{x}_{S10} + \mathbf{B}_{S10}\mathbf{u}_{S10} + \boldsymbol{\Phi}_{S10}(\mathbf{x}) - \mathbf{b}_{S10} = \mathbf{0}. \quad (120)$$

The residual $\mathbf{r}_{S10}$ is the formal object that must be closed by analysis; it is not an instruction to fabricate hardware.

Interface contract.

$$\mathcal{I}_{S10} = [\dot{m} \ P \ T \ h \ s \ \mathbf{a}_p \ \mathbf{q}'' \ \mathbf{y} \ \Sigma]^T, \quad (121)$$

where Σ is the uncertainty covariance associated with the interface variables. Every interface must provide not only a nominal value but also uncertainty and validity range.

Acceptance logic.

$$M_{stability} > M_{min}, \quad (122)$$

$$\Pr(\mathcal{F}_{S10}) < \Pr_{max}, \quad (123)$$

$$\kappa(\mathcal{O}_{S10}) < \kappa_{max}, \quad (124)$$

$$\mathcal{E}_{evidence} \geq \mathcal{E}_{required}. \quad (125)$$

Here $\mathcal{F}$ is a failure proposition, $\mathcal{O}$ is the subsystem observability map, and κ is a conditioning metric.
Evidence artifacts. Required artifacts include a model note, an uncertainty note, a hazard note, a coupling note, and a verification note. Operational procedures, component dimensions, and procurement specifications are excluded.

11.11 S11: Quantum and Optical Sensing

Physics themes: fiber gratings, spectroscopy, quantum inertial concepts, hidden-state estimation.

State subset.

$$\mathbf{x}_{S11} = [\xi_1 \ \xi_2 \ \xi_3 \ \xi_4 \ \xi_5]^T, \quad (126)$$

where the ξ_i are subsystem-specific nondimensional states: load fraction, thermal severity, stability margin, uncertainty level, and observability score.

Governing closure.

$$\mathbf{r}_{S11}(\mathbf{x}, \mathbf{u}, \boldsymbol{\theta}) = \mathbf{A}_{S11}\mathbf{x}_{S11} + \mathbf{B}_{S11}\mathbf{u}_{S11} + \boldsymbol{\Phi}_{S11}(\mathbf{x}) - \mathbf{b}_{S11} = \mathbf{0}. \quad (127)$$

The residual $\mathbf{r}_{S11}$ is the formal object that must be closed by analysis; it is not an instruction to fabricate hardware.

Interface contract.

$$\mathcal{I}_{S11} = [\dot{m} \ P \ T \ h \ s \ \mathbf{a}_p \ \mathbf{q}'' \ \mathbf{y} \ \Sigma]^T, \quad (128)$$

where Σ is the uncertainty covariance associated with the interface variables. Every interface must provide not only a nominal value but also uncertainty and validity range.

Acceptance logic.

$$M_{stability} > M_{min}, \quad (129)$$

$$\Pr(\mathcal{F}_{S11}) < \Pr_{max}, \quad (130)$$

$$\kappa(\mathcal{O}_{S11}) < \kappa_{max}, \quad (131)$$

$$\mathcal{E}_{evidence} \geq \mathcal{E}_{required}. \quad (132)$$

Here $\mathcal{F}$ is a failure proposition, $\mathcal{O}$ is the subsystem observability map, and κ is a conditioning metric.

Evidence artifacts. Required artifacts include a model note, an uncertainty note, a hazard note, a coupling note, and a verification note. Operational procedures, component dimensions, and procurement specifications are excluded.

11.12 S12: Autonomous Supervisor

Physics themes: mode governance, risk gating, uncertainty-aware control, documentation boundary enforcement.

State subset.

$$\mathbf{x}_{S12} = [\xi_1 \ \xi_2 \ \xi_3 \ \xi_4 \ \xi_5]^T, \quad (133)$$

where the ξ_i are subsystem-specific nondimensional states: load fraction, thermal severity, stability margin, uncertainty level, and observability score.

Governing closure.

$$\mathbf{r}_{S12}(\mathbf{x}, \mathbf{u}, \boldsymbol{\theta}) = \mathbf{A}_{S12}\mathbf{x}_{S12} + \mathbf{B}_{S12}\mathbf{u}_{S12} + \boldsymbol{\Phi}_{S12}(\mathbf{x}) - \mathbf{b}_{S12} = \mathbf{0}. \quad (134)$$

The residual $\mathbf{r}_{S12}$ is the formal object that must be closed by analysis; it is not an instruction to fabricate hardware.

Interface contract.

$$\mathcal{I}_{S12} = [\dot{m} \ P \ T \ h \ s \ \mathbf{a}_p \ \mathbf{q}'' \ \mathbf{y} \ \Sigma]^T, \quad (135)$$

where Σ is the uncertainty covariance associated with the interface variables. Every interface must provide not only a nominal value but also uncertainty and validity range.

Acceptance logic.

$$M_{stability} > M_{min}, \quad (136)$$

$$\Pr(\mathcal{F}_{S12}) < \Pr_{max}, \quad (137)$$

$$\kappa(\mathcal{O}_{S12}) < \kappa_{max}, \quad (138)$$

$$\mathcal{E}_{evidence} \geq \mathcal{E}_{required}. \quad (139)$$

Here $\mathcal{F}$ is a failure proposition, $\mathcal{O}$ is the subsystem observability map, and κ is a conditioning metric.

Evidence artifacts. Required artifacts include a model note, an uncertainty note, a hazard note, a coupling note, and a verification note. Operational procedures, component dimensions, and procurement specifications are excluded.

12 Interface Control Document Template

12.1 Canonical Interface Variables

Every interface is represented as a typed bundle:

$$\mathcal{B} = (\mathbf{v}_{nom}, \mathbf{U}, \mathcal{D}_{valid}, \mathcal{A}, \mathcal{H}, \mathcal{S}), \quad (140)$$

where $\mathbf{v}_{nom}$ are nominal symbolic variables, $\mathbf{U}$ is uncertainty, $\mathcal{D}_{valid}$ is the validity domain, $\mathcal{A}$ is assumption set, $\mathcal{H}$ is hazard annotation, and $\mathcal{S}$ is source/evidence class.

Interface	Variables	Verification question
Preburner to turbine	$\dot{m}, h_0, P_t, T_t, Y_k$	Does turbine power close with explicit thermal and chemical uncertainty?
Pump to injector	$\dot{m}, P, T, \rho, \sigma_{cav}$	Is the feed state inside the model's validity domain?
Injector to chamber	$\lambda_m, \phi, \tau_{mix}, \Delta P/P$	Is mixing fast enough for the selected combustion mode?
Chamber to wall	$q'', P_c, T_g, \chi_i, I_\lambda$	Does the wall stack close heat balance conservatively?
Nozzle to MHD system	$M, p, T, \sigma_e, \mathbf{B}$	Is plasma coupling bounded and energetically consistent?
Sensors to twin	$\mathbf{y}, \mathbf{R}, f_s, \tau_d$	Is the hidden state observable with acceptable delay?

13 Verification and Validation Matrix

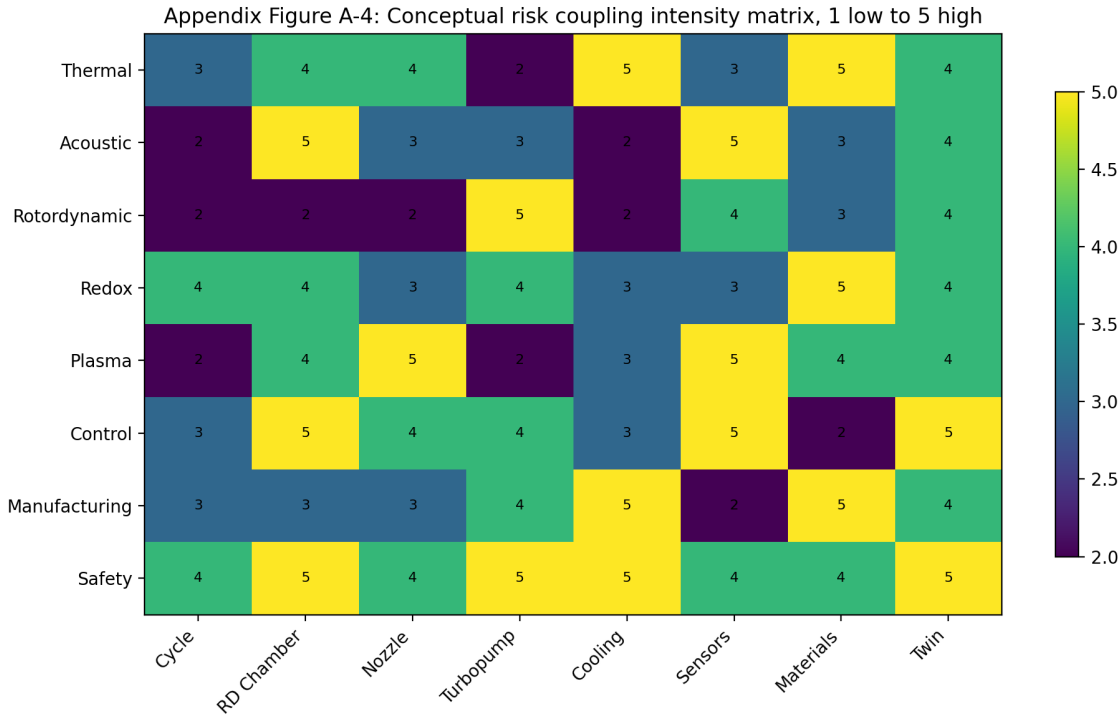


Figure 4: Risk coupling matrix for prioritizing model review.

13.1 Evidence Classes

Evidence class	Meaning
E0	Conceptual analogy only; allowed for speculative appendices.
E1	Dimensional analysis and conservation-law consistency.
E2	Reduced-order simulation with uncertainty bounds.
E3	High-fidelity simulation or literature-bounded surrogate model.
E4	Non-operational laboratory analogy unrelated to a complete engine.
E5	Independent review closure of assumptions, equations, and safety boundary.

13.2 Requirement Closure Equation

A requirement R_i is considered closed on paper if

$$C(R_i) = \mathbb{K} [E_i \geq E_{req,i}] \mathbb{K} [U_i \leq U_{max,i}] \mathbb{K} [H_i \leq H_{max,i}] \mathbb{K} [B_i = 1], \quad (141)$$

where B_i confirms the non-operational documentation boundary.

13.3 Traceability Matrix

ID	Requirement	Model artifact	Review evidence
R-01	State the cycle closure without build parameters.	Cycle residual model.	Conservation audit.
R-02	Represent detonation mode as a bounded oscillator.	RD state-space model.	Stability map review.

R-03	Bound thermal load paths with uncertainty.	Wall-stack thermal model.	Heat-balance audit.
R-04	Show electromagnetic terms conserve energy.	MHD nozzle balance.	Power audit.
R-05	Make hidden states observable.	Sensor/twin observability matrix.	Rank and conditioning check.
R-06	Maintain safety boundary.	Scope classifier and redaction rules.	Manual review.
R-07	Avoid operational specificity.	Abstract variables only.	Safety edit pass.
R-08	Capture material compatibility by region.	Redox materials map.	Coupling review.
R-09	Identify dominant coupled risks.	Risk matrix.	Review board ranking.
R-10	Preserve the spectacle.	Concept art and narrative architecture.	Coherence review.

14 FMEA-Style Paper Hazard Register

Hazard theme	Failure proposition	Paper indicator	Mitigation in this appendix
Combustion instability	Acoustic energy grows faster than damping.	$\text{Re}(\lambda) > 0$ for modal system.	Metamaterial damping model and mode map.
Detonation loss	RD wave extinguishes or multiplies unpredictably.	N_w jumps outside allowed set.	State-machine analysis only, no operating sequence.
Thermal runaway	Wall heat exceeds modeled absorption.	$\mathcal{T}_{run} \geq 0$.	Conservative heat-balance and uncertainty margin.
Rotor instability	Magnetic support loses stability.	Closed-loop eigenvalue crosses right half-plane.	Rotor model, no bearing construction recipe.
MHD inconsistency	Electromagnetic power exceeds available energy.	$\eta_{MHD} > 1$ or violates balance.	Energy audit.
Material incompatibility	Redox or fatigue penalty dominates.	Π_R or Π_F above bound.	Redox-aware zoning.
Observability failure	Hidden state cannot be inferred.	Rank loss in $\mathcal{O}$.	Sensor redundancy and twin covariance.
Scope creep	Document becomes operational.	Presence of forbidden content class.	Boundary review and redaction.

14.1 Risk Priority Number

A paper RPN is

$$RPN_i = S_i O_i D_i K_i, \quad (142)$$

where S is severity, O occurrence plausibility, D detection difficulty, and K knowledge deficit. Unlike hardware FMEA, occurrence is interpreted as conceptual plausibility under uncertainty, not expected field rate.

15 Simulation Stack Implementation

15.1 Model Layers

The recommended paper stack contains:

1. **Layer 0:** symbolic conservation equations and nondimensional groups.
2. **Layer 1:** algebraic cycle closure and steady-state residuals.
3. **Layer 2:** transient reduced-order models for chamber, pumps, walls, and controller.
4. **Layer 3:** high-fidelity submodels for RD chamber, cooling wall, MHD nozzle, and rotor dynamics.
5. **Layer 4:** uncertainty propagation, sensitivity analysis, and requirement closure.

15.2 Coupled Residual Form

The entire paper engine is a residual problem:

$$\mathbf{R}(\mathbf{x}, \dot{\mathbf{x}}, \mathbf{u}, \boldsymbol{\theta}, t) = \begin{bmatrix} \mathbf{R}_{cycle} \\ \mathbf{R}_{chem} \\ \mathbf{R}_{RD} \\ \mathbf{R}_{thermal} \\ \mathbf{R}_{rotor} \\ \mathbf{R}_{MHD} \\ \mathbf{R}_{control} \\ \mathbf{R}_{material} \end{bmatrix} = \mathbf{0}. \quad (143)$$

The coupled Jacobian

$$\mathbf{J} = \frac{\partial \mathbf{R}}{\partial \mathbf{x}} \quad (144)$$

is used not for operation, but to identify which physics dominate uncertainty and instability.

15.3 Sensitivity and Sobol Indices

For a scalar output $Y = f(\boldsymbol{\theta})$, variance decomposes as

$$\text{Var}(Y) = \sum_i V_i + \sum_{i < j} V_{ij} + \cdots, \quad (145)$$

with first-order sensitivity

$$S_i = \frac{V_i}{\text{Var}(Y)}. \quad (146)$$

High S_i terms become research priorities.

16 Paper Maturity Roadmap

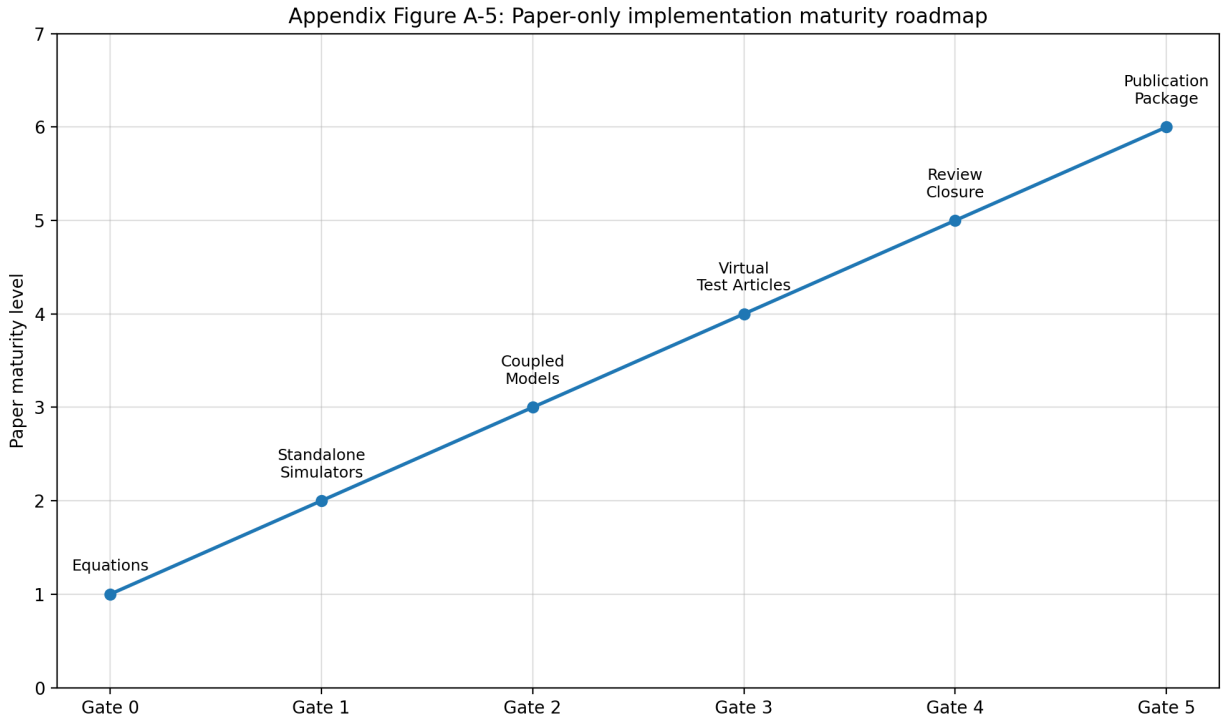


Figure 5: Maturity is defined by analysis closure, not hardware readiness.

16.1 Gate Definitions

Gate	Name	Exit criteria
G0	Concept Freeze	Architecture, safety boundary, and subsystem list complete.
G1	Equation Freeze	Governing equations and variables defined; no untyped states.
G2	Interface Freeze	Every subsystem has input/output bundles and uncertainty.
G3	Coupled Model Freeze	Integrated residual model closes symbolically.
G4	Verification Freeze	Requirement matrix, FMEA, and evidence classes reviewed.
G5	Publication Freeze	Document is complete, non-operational, and internally consistent.

16.2 Research Debt Ledger

Research debt is tracked as

$$D_{research} = \sum_i w_i (1 - E_i/E_{target,i})_+ + \sum_j \nu_j U_j, \quad (147)$$

where E_i is evidence maturity and U_j is unresolved uncertainty.

17 Detailed Requirement Catalogue

ID	Requirement statement	Verification method
A-001	Thermal requirement: the paper model shall define a bounded, auditable, uncertainty-tagged relationship for the relevant subsystem without introducing actionable build or operation parameters.	Equation review, interface review, evidence class audit.
A-002	Combustion requirement: the paper model shall define a bounded, auditable, uncertainty-tagged relationship for the relevant subsystem without introducing actionable build or operation parameters.	Equation review, interface review, evidence class audit.
A-003	MHD requirement: the paper model shall define a bounded, auditable, uncertainty-tagged relationship for the relevant subsystem without introducing actionable build or operation parameters.	Equation review, interface review, evidence class audit.
A-004	Materials requirement: the paper model shall define a bounded, auditable, uncertainty-tagged relationship for the relevant subsystem without introducing actionable build or operation parameters.	Equation review, interface review, evidence class audit.
A-005	Controls requirement: the paper model shall define a bounded, auditable, uncertainty-tagged relationship for the relevant subsystem without introducing actionable build or operation parameters.	Equation review, interface review, evidence class audit.
A-006	Sensors requirement: the paper model shall define a bounded, auditable, uncertainty-tagged relationship for the relevant subsystem without introducing actionable build or operation parameters.	Equation review, interface review, evidence class audit.
A-007	Safety requirement: the paper model shall define a bounded, auditable, uncertainty-tagged relationship for the relevant subsystem without introducing actionable build or operation parameters.	Equation review, interface review, evidence class audit.
A-008	Cycle requirement: the paper model shall define a bounded, auditable, uncertainty-tagged relationship for the relevant subsystem without introducing actionable build or operation parameters.	Equation review, interface review, evidence class audit.
A-009	Thermal requirement: the paper model shall define a bounded, auditable, uncertainty-tagged relationship for the relevant subsystem without introducing actionable build or operation parameters.	Equation review, interface review, evidence class audit.
A-010	Combustion requirement: the paper model shall define a bounded, auditable, uncertainty-tagged relationship for the relevant subsystem without introducing actionable build or operation parameters.	Equation review, interface review, evidence class audit.
A-011	MHD requirement: the paper model shall define a bounded, auditable, uncertainty-tagged relationship for the relevant subsystem without introducing actionable build or operation parameters.	Equation review, interface review, evidence class audit.
A-012	Materials requirement: the paper model shall define a bounded, auditable, uncertainty-tagged relationship for the relevant subsystem without introducing actionable build or operation parameters.	Equation review, interface review, evidence class audit.
A-013	Controls requirement: the paper model shall define a bounded, auditable, uncertainty-tagged relationship for the relevant subsystem without introducing actionable build or operation parameters.	Equation review, interface review, evidence class audit.

A-014	Sensors requirement: the paper model shall define a bounded, auditable, uncertainty-tagged relationship for the relevant subsystem without introducing actionable build or operation parameters.	Equation review, interface review, evidence class audit.
A-015	Safety requirement: the paper model shall define a bounded, auditable, uncertainty-tagged relationship for the relevant subsystem without introducing actionable build or operation parameters.	Equation review, interface review, evidence class audit.
A-016	Cycle requirement: the paper model shall define a bounded, auditable, uncertainty-tagged relationship for the relevant subsystem without introducing actionable build or operation parameters.	Equation review, interface review, evidence class audit.
A-017	Thermal requirement: the paper model shall define a bounded, auditable, uncertainty-tagged relationship for the relevant subsystem without introducing actionable build or operation parameters.	Equation review, interface review, evidence class audit.
A-018	Combustion requirement: the paper model shall define a bounded, auditable, uncertainty-tagged relationship for the relevant subsystem without introducing actionable build or operation parameters.	Equation review, interface review, evidence class audit.
A-019	MHD requirement: the paper model shall define a bounded, auditable, uncertainty-tagged relationship for the relevant subsystem without introducing actionable build or operation parameters.	Equation review, interface review, evidence class audit.
A-020	Materials requirement: the paper model shall define a bounded, auditable, uncertainty-tagged relationship for the relevant subsystem without introducing actionable build or operation parameters.	Equation review, interface review, evidence class audit.
A-021	Controls requirement: the paper model shall define a bounded, auditable, uncertainty-tagged relationship for the relevant subsystem without introducing actionable build or operation parameters.	Equation review, interface review, evidence class audit.
A-022	Sensors requirement: the paper model shall define a bounded, auditable, uncertainty-tagged relationship for the relevant subsystem without introducing actionable build or operation parameters.	Equation review, interface review, evidence class audit.
A-023	Safety requirement: the paper model shall define a bounded, auditable, uncertainty-tagged relationship for the relevant subsystem without introducing actionable build or operation parameters.	Equation review, interface review, evidence class audit.
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18 Mathematical Appendix: Nondimensional Groups

Group	Definition	Interpretation
Mach	$M = v/a$	Compressibility and nozzle expansion state.
Reynolds	$Re = \rho v L / \mu$	Ratio of inertial to viscous transport.
Damkohler	$Da = \tau_{flow} / \tau_{chem}$	Chemical completion relative to residence.
Karlovitz	$Ka = \tau_{flame} / \tau_{\eta}$	Turbulence-chemistry interaction marker.
Peclet	$Pe = Re Pr$	Advective to diffusive heat transport.
Biot	$Bi = h L / k$	Internal to external thermal resistance.
Hartmann	$Ha = BL \sqrt{\sigma_e / \mu}$	Magnetic to viscous force scale.
Magnetic Reynolds	$Rm = \mu_0 \sigma_e v L$	Magnetic advection relative to diffusion.
Stuart	$N = \sigma_e B^2 L / (\rho v)$	Lorentz to inertial force scale.
Strouhal	$St = f L / v$	Unsteady oscillation scaling.
Knudsen	$Kn = \lambda / L$	Continuum validity marker.
Cavitation	$\sigma_c = (p - p_v) / (0.5 \rho v^2)$	Liquid pump cavitation margin surrogate.

A paper model should never hide behind dimensional variables when nondimensional structure reveals the true regime. The nondimensional state vector

$$\mathbf{\Pi} = [M \ Re \ Da \ Ka \ Pe \ Bi \ Ha \ Rm \ N \ St \ Kn \ \sigma_c]^T \quad (148)$$

is a concise language for multidisciplinary coupling.

19 Manual Governance and Redaction Checklist

19.1 Safety Redaction Rules

Before release, the document is reviewed against the following content classes:

1. Remove exact geometries that would enable fabrication.
2. Remove procedural sequences that would enable operation.
3. Remove procurement-specific component lists.
4. Remove exact propellant loading, pressurization, or ignition details.
5. Remove test-stand configuration instructions.
6. Preserve equations, abstract models, qualitative architecture, and safety boundaries.

19.2 Final Paper Release Criteria

$$Release = \prod_{i=1}^{N_R} C(R_i) \prod_{j=1}^{N_H} \mathbb{I}[RPN_j < RPN_{max}] \prod_{k=1}^{N_B} \mathbb{I}[B_k = 1]. \quad (149)$$

A single failed boundary condition blocks release.

19.3 Closing Statement

Chimera Prometheus is implemented when it becomes a coherent multidisciplinary model: a set of equations that conserve, interfaces that trace, sensors that observe, materials that explain their survival, and a digital twin that admits uncertainty. Its grandeur comes from refusing to be merely an engine. On paper, it is a thermodynamic organism, a plasma instrument, a materials experiment, a nonlinear oscillator, and a manifesto for the physics of propulsion beyond the visible plumbing era.